

FAKHIRAH SHAKILA BINTI KHAIRUL ARIFFIN

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EDUCATION

UNIVERSITI TEKNOLOGI MALAYSIA (UTM)

Bachelor Of Computer Science (Graphics and Multimedia Software) With Honours

CGPA: 3.44

Relevant Coursework: Object-Oriented Programming, Programming Techniques, Data Structures & Algorithms, Software Engineering, Mathematics for Computer Graphics, Computer Graphics, Geometric Modelling, Human-Computer Interaction, and Virtual Reality & Simulation.

Skudai, Johor

Oct 2022 – July 2026

EXPERIENCE

SOFTWARE ENGINEER | AR Mechatronics PLT

Sept 2026 – Feb 2026

AR Mechatronics PLT is a technology startup and UTM spin-off company based in Skudai, Johor Bahru, Malaysia. Founded on June 1, 2019, it specializes in artificial intelligence (AI), the Internet of Things (IoT), robotics, and smart community software solutions like the Resident Easy Decision System (MYREDS).

- Working on a Raspberry Pi-based Tender Box automation system using Python, Django, Flask, and GPIO.
- Developed API communication between web and hardware systems, implemented scheduling and override functions, integrated solenoid, sensors and cameras, debugged hardware/software issues, and collaborated using Git/Bitbucket.
- Created system documentation and improved Tkinter/Django interfaces.

PERSONAL AND UNIVERSITY PROJECTS

HOLOLINGUA: AI-POWERED MULTILINGUAL TUTOR FOR INTERACTIVE LANGUAGE LEARNING IN VR (FINAL YEAR PROJECT) | Unity, OpenXR, Convai SDK

- Developed virtual reality language learning application featuring AI-driven conversational tutors powered by the Convai SDK.
- Optimized application builds and rendering pipelines for standalone VR hardware to ensure smooth performance during interactive exercises.

MULTIPLAYER VR FOOD TRUCK SIMULATION | Godot, Blockbench, Blender.

Collaborative VR game development focusing on 3D asset modelling, instruction UI design, and environment audio.

MAPPY X NIER (FANMADE CROSSOVER) | Defold, Lua, 2D Animation

- Designed and programmed a 2D retro-style action-platformer combining the arcade mechanics of Mappy with the thematic aesthetic of Nier: Automata.
- Constructed custom 2B character animation frames, sprite loops, and promotional cover assets.
- Implemented randomized item-spawning logic and AI pathfinding algorithms for enemy units.

“I DON’T WANNA BE ITIK SALAI” | PlayCanvas, Yuka.js, JavaScript

- Engineered a psychological horror-chaser web game focusing on high-tension NPC pursuit mechanics.
- Integrated Yuka.js for autonomous NPC navigation and utilized raycasting line detection to simulate accurate enemy line-of-sight and player detection.

LEAP MOTION PORTAL GAME | Unity, Leap Motion SDK, C#

- Developed an interactive portal game utilizing Leap Motion hardware for real-time hand tracking and gesture recognition.
- Implemented gesture-triggered event systems to dynamically instantiate portals based on specific user hand signs.

SHINGEKI NO ALIEN | GDevelop, 3D Rendering

- Built a fast-paced 3D survival shooter featuring a high-stakes combat scenario.
- Configured AI navigation meshes and tracking behaviors to drive dynamic enemy movement.

MBPG SMART TENDER BOX SYSTEM | AR Mechatronics PLT (Internship) | Python, Raspberry Pi, Django, REST APIs

- Programmed and optimized Raspberry Pi control scripts using Python to enhance embedded system functionality and hardware-software integration.
- Configured GPIO connections to automate and control relay modules and solenoid physical security locks.
- Integrated physical IoT devices with a Django backend, developing custom REST APIs to process requests between the hardware and web-based servers.

HOSP-TRACKER NAVIGATION ROBOT | Raspberry Pi, Python, Antigravity IDE

- Programmed a Raspbot automated navigation system for a hospital environment as part of a Robotic Design & Development Technology with Raspberry Pi Microprocessor Certificate program.
- Designed and prototyped a companion web dashboard to track and visualize the patient journey flow alongside the robot's hardware operations.

MULTIUSER INTERACTION APPLICATION | Unity, Firebase, SENEM, C#

- Integrated a Firebase backend for secure player authentication and real-time NPC database management.
- Implemented a multiplayer economy system where users interact with NPCs and complete Q&A challenges to earn virtual currency.

VOICE CLONING DIGITAL AVATAR | Unity, Google STT, ElevenLabs API

- Developed conversational interface between users and a 3D digital avatar using Google Speech-to-Text and ElevenLabs Text-to-Speech.
- Programmed asynchronous API calls to process natural language inputs and generate real-time synthesized voice responses.

INTERACTIVE 3D CHARACTER RIG | OpenGL, GLUT, C++

Programmed a low-level graphics application allowing users to dynamically manipulate and adjust a 3D avatar's articulated body parts using spatial transformations.

SHADER-DRIVEN RENDERING WORKSHOP | WebGL, Hands-on Curriculum

Designed and presented an academic workshop teaching progressive WebGL rendering techniques and shader implementation.

3D MODELLING & SHADING WORKSHOP FACILITATOR | Blender

Designed and conducted an academic technical workshop teaching 3D sculpting, UV editing, shader configuration, and texture mapping workflows to peers.

FOOD ARCADE DINING TABLE APP | Figma, UX Research

Developed an interactive UI prototype and conducted comprehensive user testing sessions, providing task-based instructions to observe user flow, gather feedback, and iterate on design mechanics.

STUDENT REGISTRATION WEBSITE | Laravel, PHP, Git

Programmed a backend registration system utilizing the Laravel MVC framework, managing version control and collaborative branching via Git and GitHub.

WHACK-A-WORD-HAM (PUBLISHED MOBILE GAME) | Python, Kivy Library, Antigravity IDE

- Developed and self-published an interactive word puzzle game on the Google Play Store via the Google Play Console.
- Programmed core gameplay loops using the Kivy framework, requiring players to dynamically rearrange letters into meaningful words.

DATA VISUALIZATION LAB | WEKA, Kaggle Datasets

Processed and visualized complex public datasets from Kaggle using WEKA to identify data trends and classification models.

ADDITIONAL

TECHNICAL SKILLS:

- Languages: Python, C#, C++, JavaScript, PHP, Lua, GLSL (Shader Language)
- Frameworks & Libraries: Django, Laravel, Kivy, OpenGL, WebGL, GLUT, Yuka.js, OpenXR.
- Databases: Firebase, MySQL, SQLite, PostgreSQL
- Tools & Cloud:
 - Game Engines: Unity, Godot, Defold, PlayCanvas, GDevelop..
 - 3D & Design: Blender, Blockbench, Figma.
 - Development & Version Control: Git, GitHub, Antigravity IDE.
 - APIs & SDKs: Convai SDK, Leap Motion SDK, ElevenLabs API, Google STT.
 - Hardware & Deployment: Raspberry Pi, Meta Quest 3s, Google Play Console.
 - Data Analytics: WEKA, Power BI.

CERTIFICATIONS:

- Robotic Design and Development Technology with Raspberry Pi Microprocessor
| KPT Sponsored Training Program August 2026
GMP Quality Consultants | Held at Universiti Teknologi Malaysia (UTM)
 - Completed intensive technical training covering AI robotics design, electrical circuitry, microcontrollers, and robot motion control.
 - Assessed via practical project work, group assignments, and a comprehensive written examination.
- Graphic Design Essentials | Canva Design School | The Canva Team Sept 2026
 - Completed the Graphic Design Essentials course, which is the equivalent to 45 minutes of training.